# *Full core depletion study for the Fluoride High Temperature Reactor (FHR) benchmark study

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## *SCIENTIFIC ACHIEVEMENT:

Fluoride salt-cooled high temperature reactors (FHRs) and using TRISO particles for fuel are an advanced reactor concept under active and continued research interest. This reactor has similar heat transfer characteristics to MSRs like the MSRE, however it has different neutronic behavior as the fuel is in a solid state. As part of the OECD/NEA FHR benchmark, I am using OpenMC to perform reactor physics and depletion calculations on a reference model of the FHR for a phase of the benchmark, focusing on depletion. I have already obtained results for an exercise of this phase for fission density, isotopics, and k-effective. This work will continue with a subsequent exercise that splits the reactor into 15 axial depletion zones.

## *SIGNIFICANCE:

The complex structure of the TRISO particles, and the new materials used in advanced reactors like the FHR, necessitate high performance computing to study them. The OECD/NEA FHR benchmark provides a unique opportunity to have a wide ranging code-to-code comparison of a complex reactor model, serving to validate both the codes used and to provide predictive data on key reactor physics parameters such as k-effective, reacticity coefficients, and fission density. This work is alligned with both DOE-NE and NRC goals for modeling advanced non-LWRs to aid in licensing and future deployment.

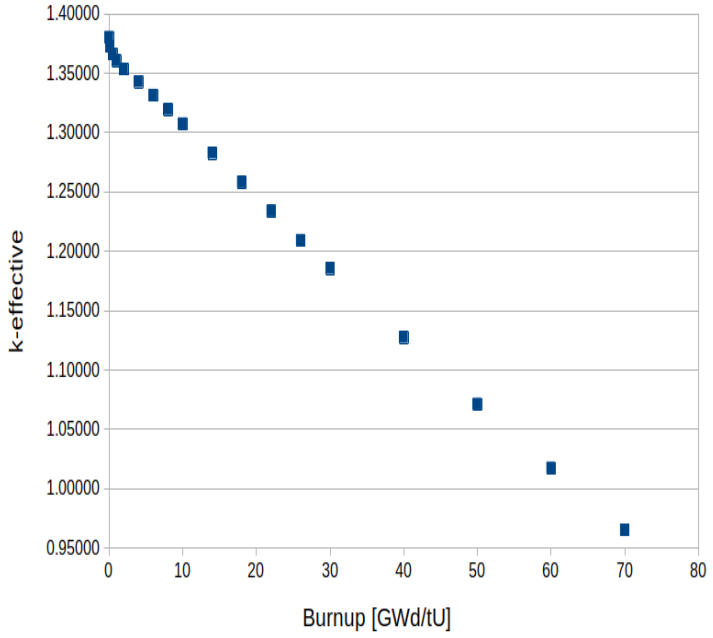
## *KEY PUBLICATIONS:

None

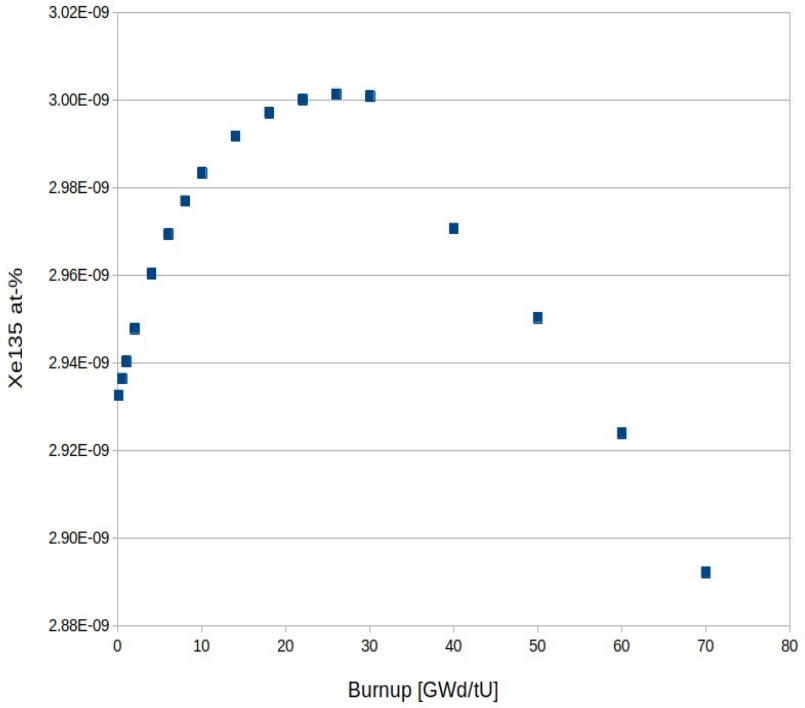
## *SPONSOR/PROGRAM:

OECD/NEA
Cody Permann

## *IMAGES AND CAPTION:




K-effective change with burnup in the FHR




Xe135 concentration change with burnup in the FHR